

IBM Employee Attrition Analysis

Data Analysis & Visualization Project

Tools Used:

- Python
- Pandas
- NumPy
- Seaborn
- Matplotlib

Dataset: IBM HR Analytics Employee Attrition Dataset.

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Topics Covered:

- Employee Demographics
Department & Job Role Analysis
Gender & Age Distribution
Attrition Rate Analysis
Business Insights & Observations

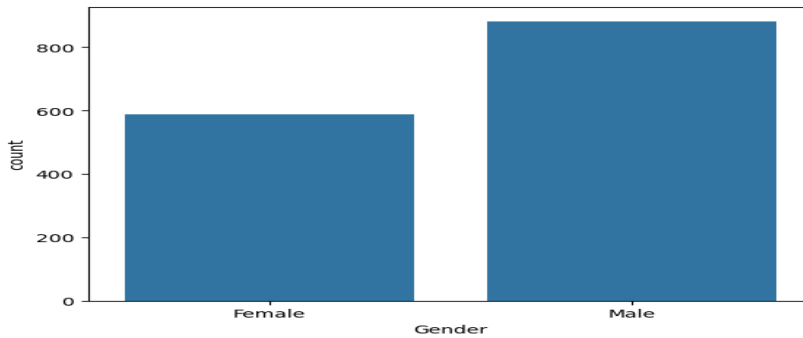
Department & Job Role-wise Attrition Rate Analysis

JobRole	Healthcare Representative	Human Resources	Laboratory Technician	Manager	Manufacturing Director	Research Director	Research Scientist	Sales Executive	Sales Representative	All
Department										
Human Resources / HR	NaN	0.230769	NaN	0.000000	NaN	NaN	NaN	NaN	NaN	0.190476
Research & Development / R&D	0.068702	NaN	0.239382	0.055556	0.068966	0.025	0.160959	NaN	NaN	0.138398
Sales	NaN	NaN	NaN	0.054054	NaN	NaN	NaN	0.174847	0.39759	0.206278
All	0.068702	0.230769	0.239382	0.049020	0.068966	0.025	0.160959	0.174847	0.39759	0.161224

Observation:

- Sales Representatives have the highest attrition rate (39.75%) among all job roles.
- Sales Executives have an attrition rate of approximately 17.45%.
- In R&D, attrition is highest among Laboratory Technicians (23.93%).
- Research Scientists show around 16.09% attrition.
- Human Resources has an overall attrition rate of approximately 19.04%.
- Sales department has the highest overall attrition rate at 20.63%.
- R&D department has the lowest overall attrition rate at approximately 13.83%.
- The analysis indicates that sales-related roles experience comparatively higher employee turnover.

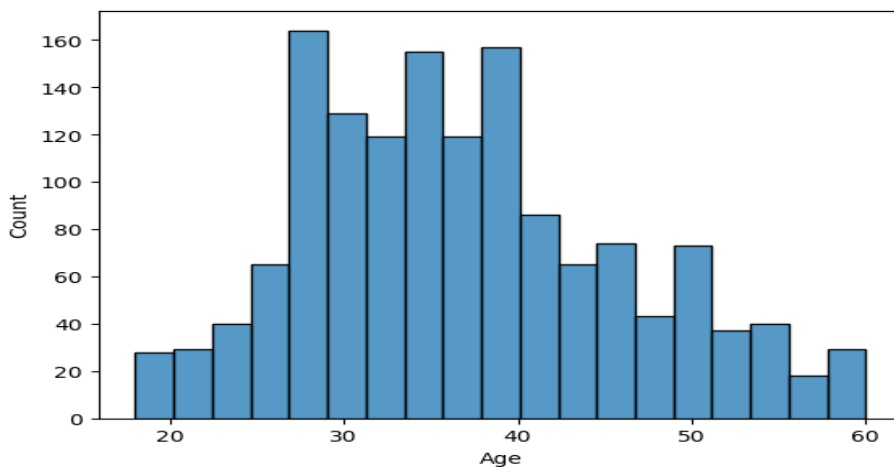
Gender Distribution of Employees – Bar Plot



Observation:

- The male employees are higher in number than female employees.
- There are approximately 850 male employees and 600 female employees.
- The workforce shows a male-dominated distribution.
- Female employees represent a smaller proportion of the total workforce.
- The difference suggests that gender representation is not evenly distributed in the organisation.

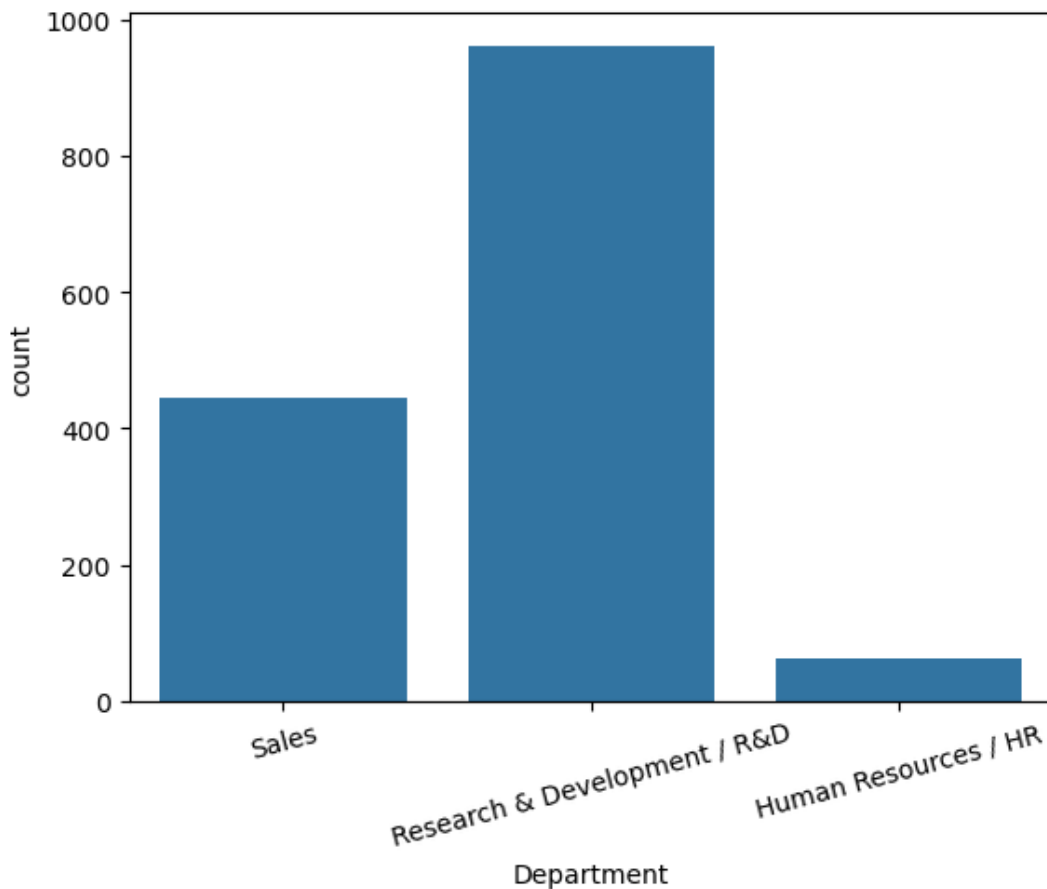
Employee Age Distribution – Histogram



Observation:

- Employees are distributed across an age range of approximately 18 to 60 years.
- The highest concentration of employees is between 27 and 40 years.
- The 27–28 age group has the highest frequency.
- Employee count generally decreases after the age of 40 years.
- Very few employees are in the 50–60 years age range.
- Overall, the workforce is relatively young, with most employees below 40 years.

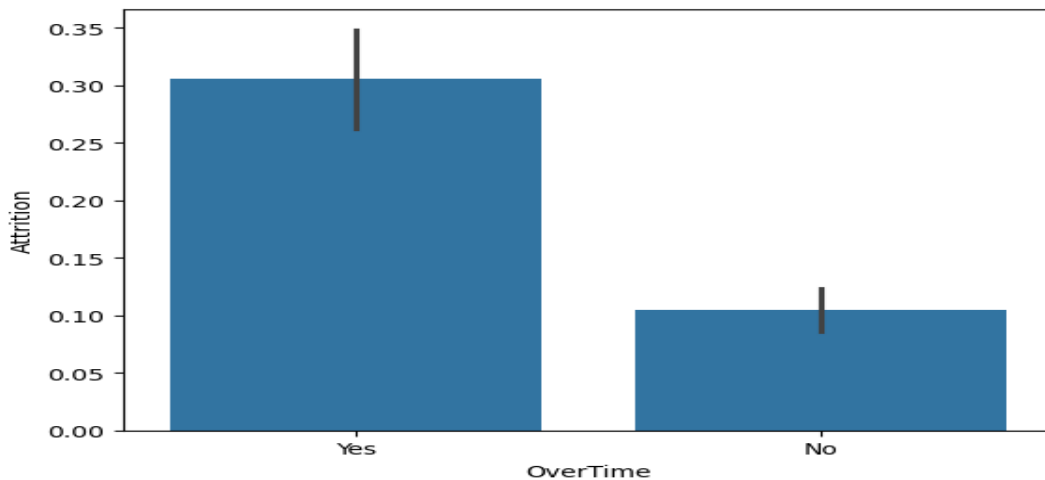
Employee Distribution by Department – Bar Plot



Observation:

- R&D has the highest number of employees, with nearly 1,000 employees.
- Sales has approximately 440 employees, making it the second-largest department.
- HR has the lowest workforce, with around 60 employees.
- R&D has more than twice the number of employees in Sales.
- The employee distribution is highly uneven across departments.
- R&D accounts for the majority of the organisation's workforce.
- HR represents only a small portion of the total employees.
- The organisation appears to have a strong focus on Research & Development activities.

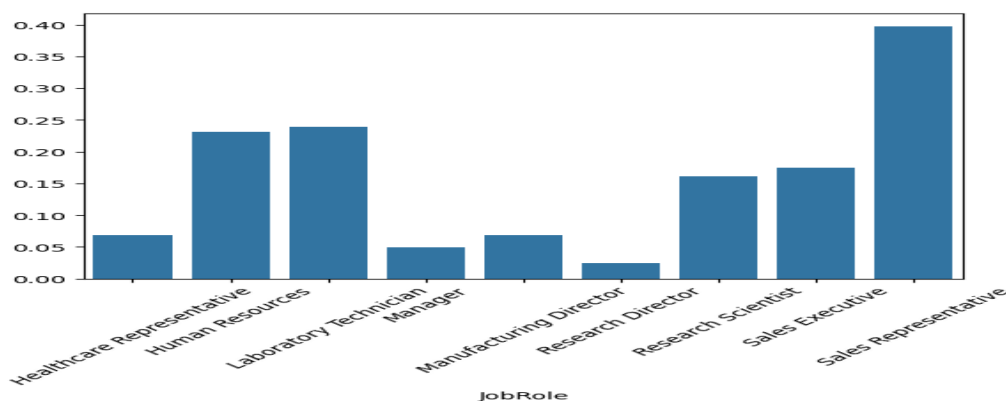
Attrition Rate by Gender – Bar Plot



Observation:

- Female employees have a higher attrition rate of approximately 30.8%.
- Male employees have a lower attrition rate of around 17%.
- Female attrition is significantly higher than male attrition.
- The difference indicates that gender may be associated with employee attrition.
- The organisation should further investigate the possible reasons for higher female employee turnover.

Attrition Rate by Job Role – Bar Plot



Observation:

- Sales Representatives have the highest attrition rate ($\approx 40\%$).
- Laboratory Technicians and Human Resources have relatively high attrition rates of around 23–24%.
- Sales Executives have an attrition rate of approximately 18%.
- Research Scientists show around 16% attrition.
- Manufacturing Directors have a moderate attrition rate of about 7%.
- Healthcare Representatives have around 7% attrition.
- Managers have a low attrition rate of approximately 5%.
- Research Directors have the lowest attrition rate ($\approx 3\%$).
- Overall, sales and some operational roles show higher employee turnover, while senior roles have comparatively lower attrition.